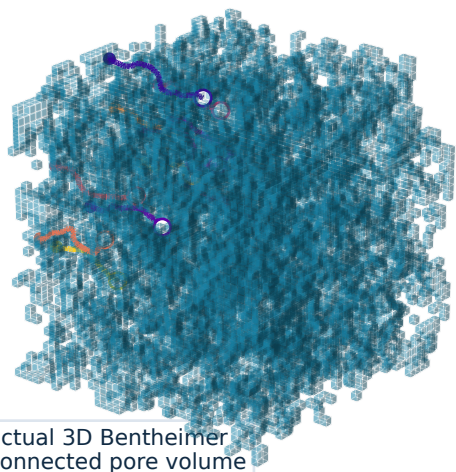


The full memory exists before the model forgets

A resolved pore-scale simulation contains geometry, velocity history, path order, and particle neighborhoods.

a memory-rich pore world



actual 3D Bentheimer
connected pore volume

b trajectory histories



fast channel
velocity history



slow/tortuous path
path order



pair divergence
neighborhood loss



near encounter
reaction opportunity

c what reduced models discard



reduced model

cannot keep
the whole history

**which forgetting
is safe?**